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ABAP Conversions

Deloitte Technology Academy (DTA)

Agenda

Topic	Descriptions	Duration
Data Transfer	Types of Data, Data Transfer Into SAP System, Data Transfer Using Interfaces, Types of Data Transfer, Data Analysis, Data Transfer Methods, and Execution of Data Transfer	8 hours
Working With Files	Overview, File Processing, Close/Delete Files, Maintain and Use Logical File Names, Create Logical Files, Local Sequential Files, and Keywords	

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Learning Objectives

By the end of this session, you will be able to:

- Describe types of data and how to transfer data
- Explain data analysis and data transfer methods
- Describe the execution of the data transfer
- Describe file transfer
- Transfer data from a presentation server to an application server and vice-versa
- Discuss the usage of logical file names
- Perform operations on files on application servers

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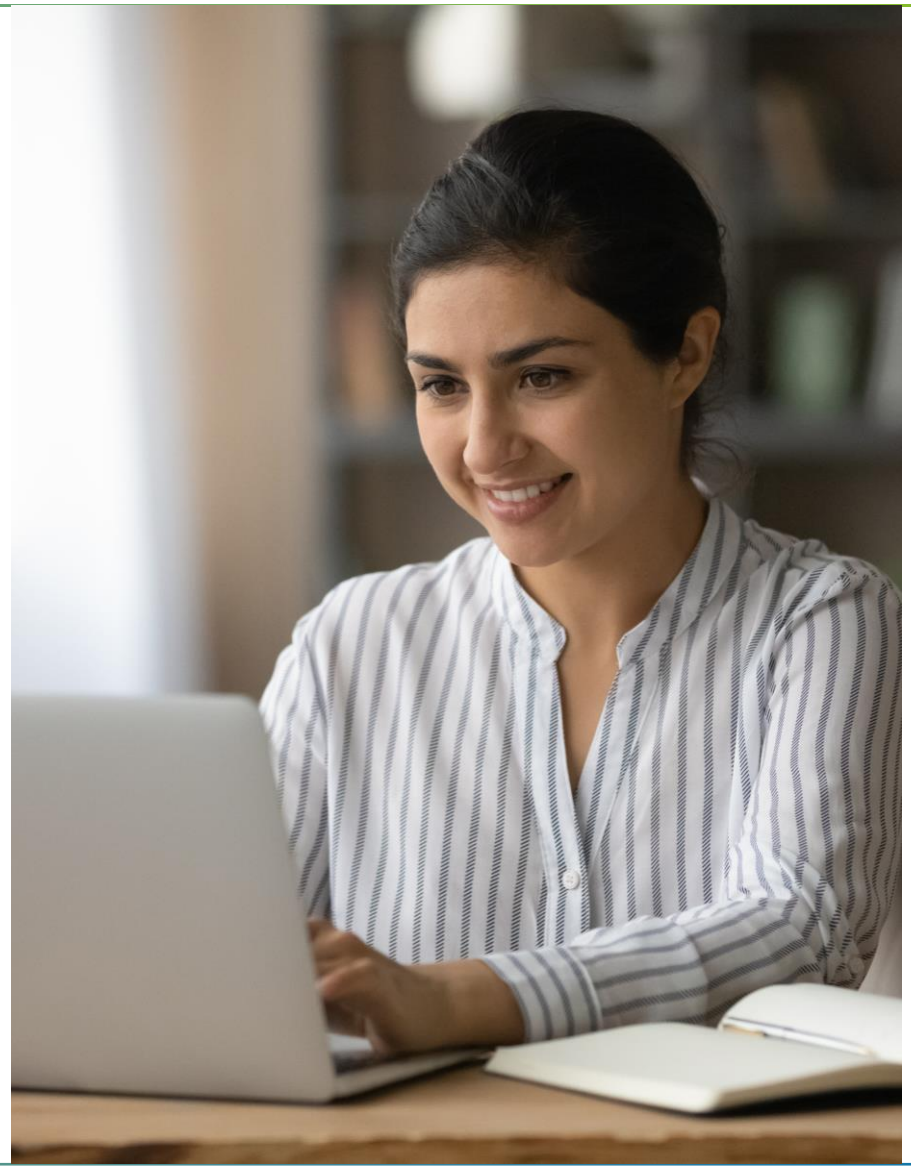
Data Transfer



Main Business Scenario

- When a company installs an R/3 system, the data from the legacy system is to be transferred to the new system.
- Customer data, Financial Accounting (FI) documents, and assets are to be transferred.

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Types of Data

There are two types of data

- Master data
- Transaction data

Master Data	Transaction Data
Customer data	Sales order
Vendor data	Purchase orders
GL accounts	Accounting documents
Materials	

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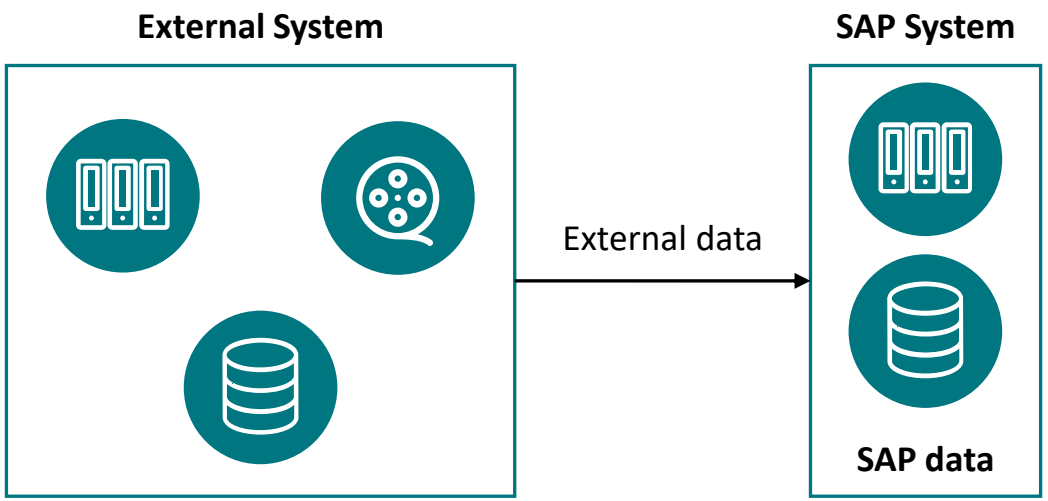
Data Transfer Into SAP System

- A large amount of data cannot be transferred manually from an external system to the R/3 system for economic reasons.
- A data transfer is required, which transfers the data in the background automatically.
- A transfer is required, for example, when:
  - A new SAP system is installed and data from another system must be transferred
  - Data is transferred from external systems into the SAP system on a regular basis. For instance, if data in some areas of the company is created in external systems and needs to be integrated into the SAP system
- The data must be transported into the R/3 system in some way.



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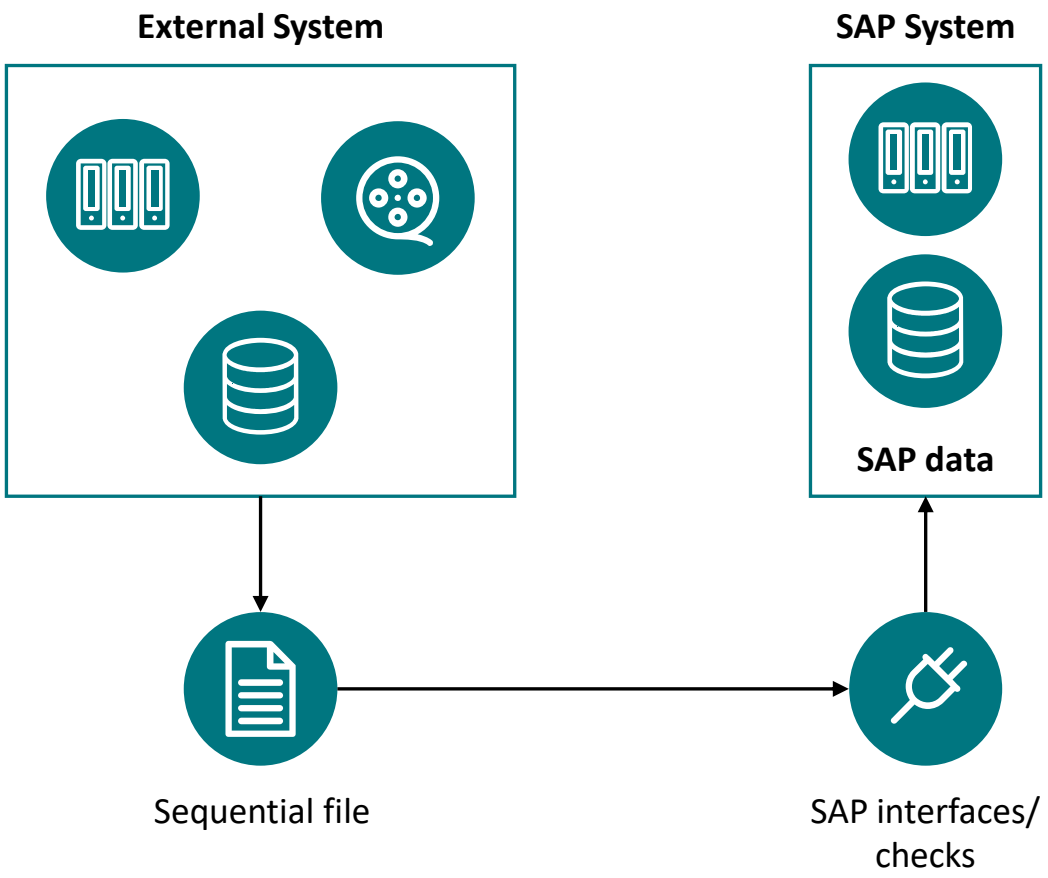
Data Transfer Using Interfaces

- Data integrity must be guaranteed when transforming data from another SAP system or from an external system.
- The same checks that are used online must be carried out for the data transfer.
- As the online checks are very extensive in the transactions and are partly cross-application, it is very difficult to carry these out yourself.
- SAP provides special interfaces for data transfer. These partly use the transactions and their checks to transfer data into the SAP system.



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Types of Data Transfer

There are two types of data transfer

- Conversion
- Interface

Conversion	Interface
Generally, used for master data load.	Generally used for transaction data transfers.
Data that needs to be transferred only once.	Transfer data periodically from external systems to SAP and vice versa.

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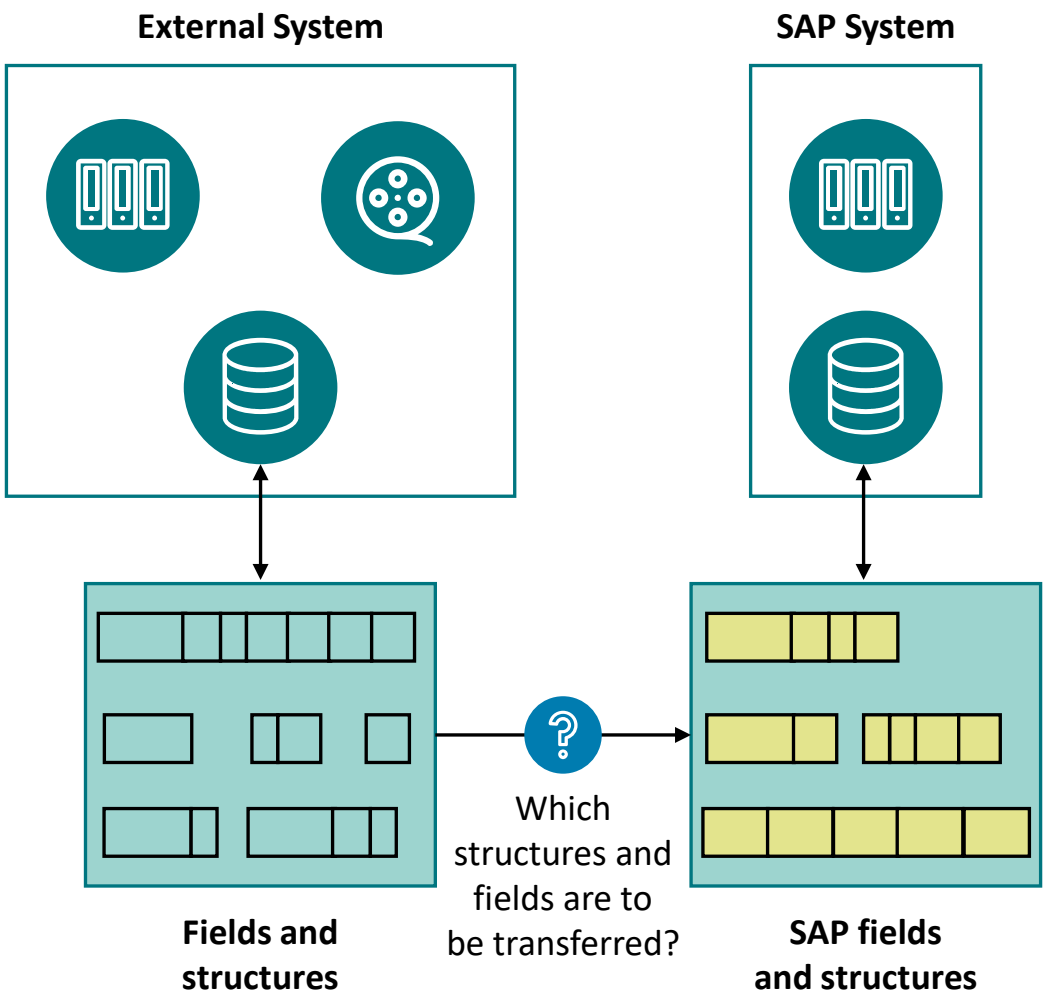
Data Analysis

- The first step is to analyze the data, which is to be transferred into the SAP system.
- The objective is to specify how existing data should be mapped to SAP data structures.
- You must determine which structures contain the data, as well as which applications and transactions are typically used to transfer data online into the SAP system.



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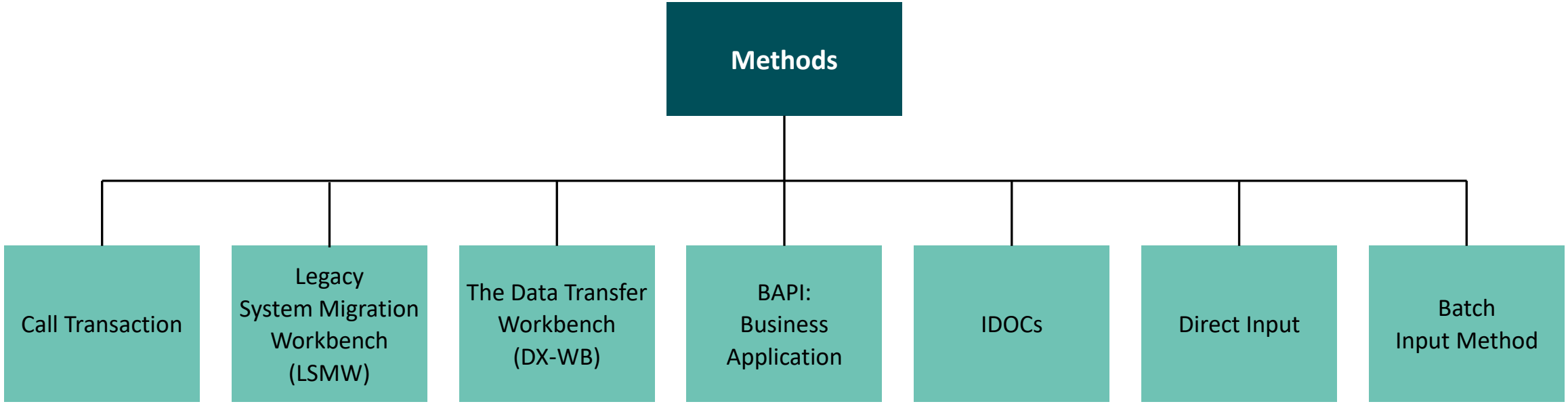
Data Transfer Methods



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Execution of Data Transfer

To execute a data transfer, you must understand the following:

- The transaction used to transfer data to the SAP system
- How to work with sequential files?
- The functions of various data transfer methods
- ABAP programming

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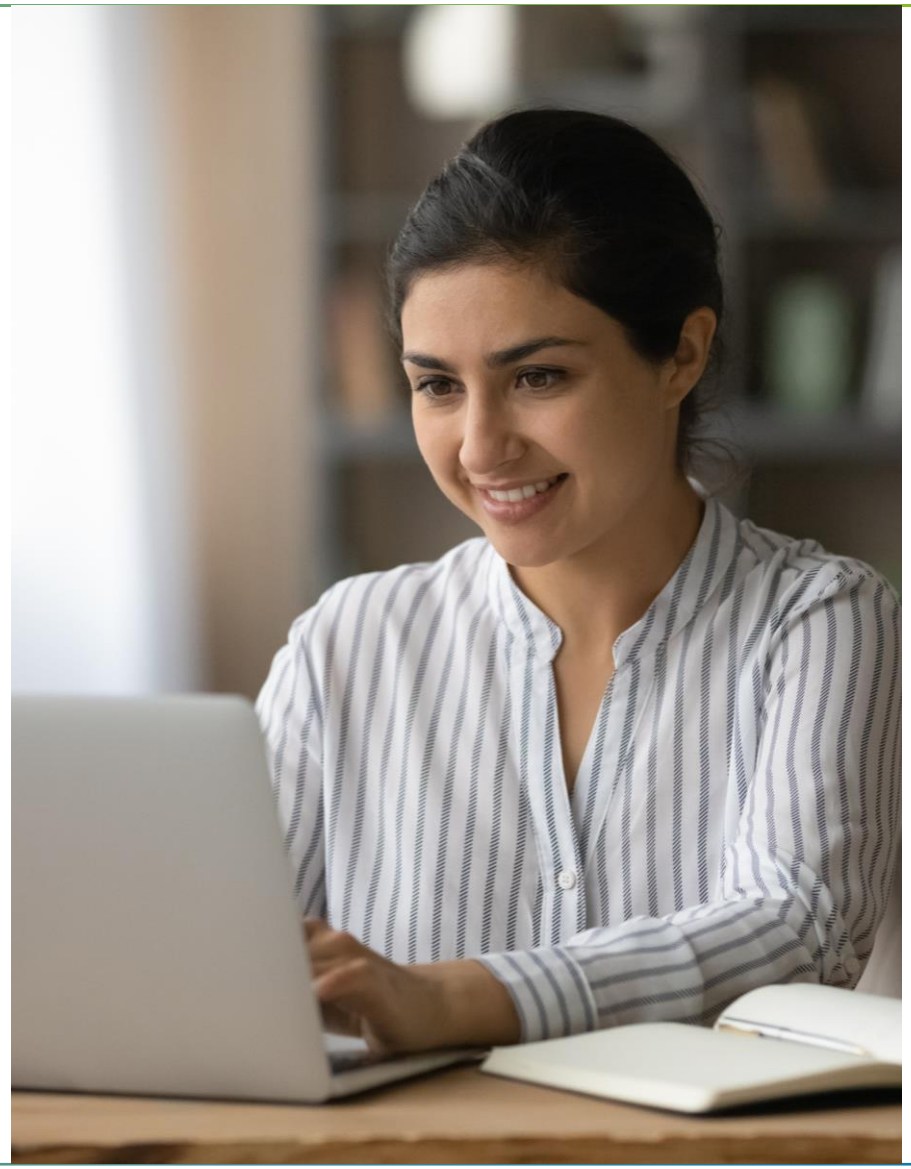
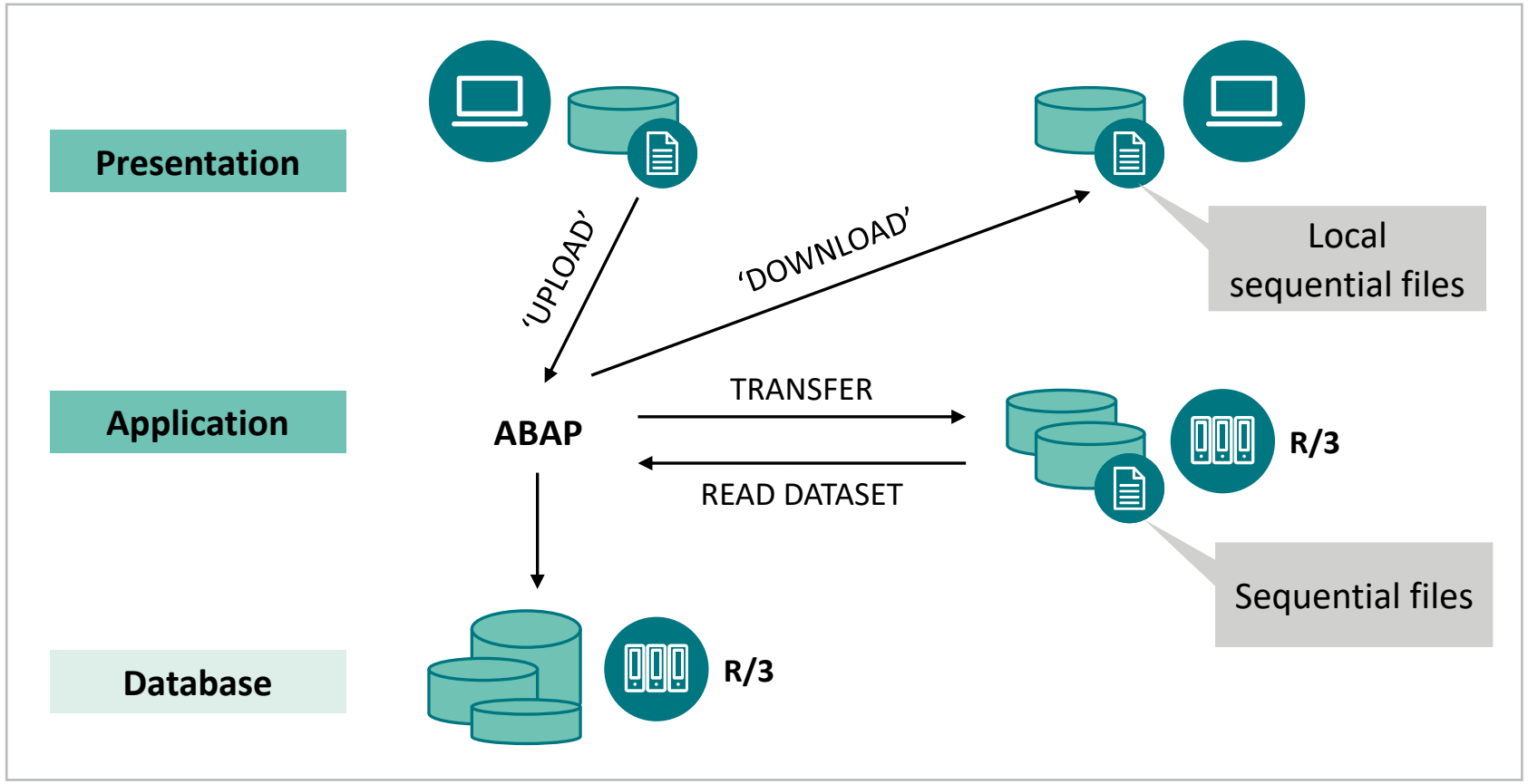
File Transfer



File Transfer

The file transfer between the three layers is given below.



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Overview—File Transfer

The SAP system comprises three distinct layers:

Database	The database system resides on this layer.
Application (Operating System)	The main process takes place on this layer (i.e., ABAP programs run on this layer).
Presentation	The “front end” to the SAP system (i.e., SAPGUI) resides on this layer.

- These three layers can be arranged in many technical architectures. For the remainder of this section, assume that these three layers reside on separate computers called “servers.”
- In this chapter, we will learn how to process sequential files on an application server. These files are called “datasets.”
- The concept of processing sequential files is important because these files are used to transfer data between an SAP system and an external system.

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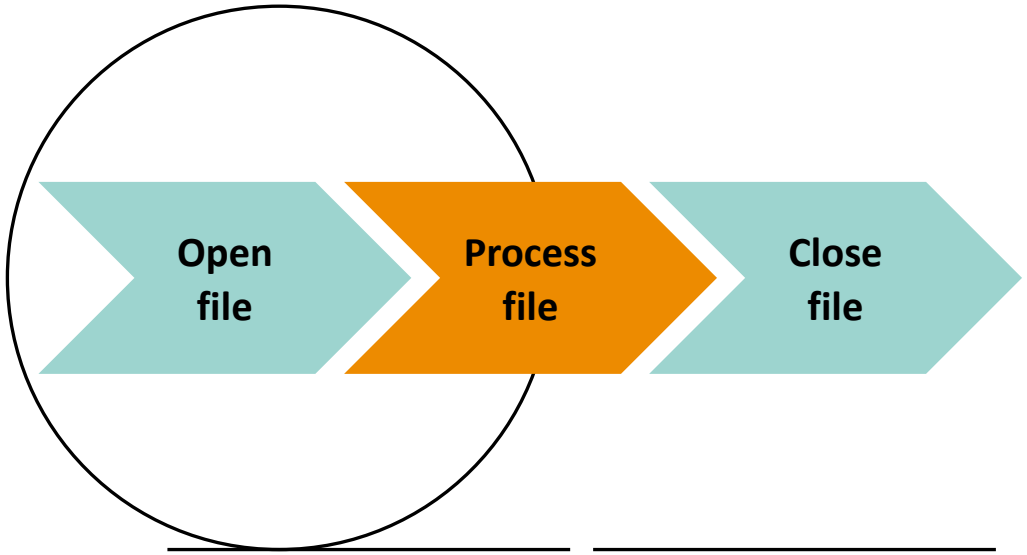
File Processing

- “Processing” a file refers to either creating a file or reading a file.
- Before a sequential file can be processed, it must be opened.
- At most, six files can be opened at the same time in each session.
- After a sequential file has been processed, it must be closed.
- In the following slides, we will learn the ABAP statements dealing with sequential files on the application server.
- Each of these statements needs to have a file name specified. This file name includes the full directory path and the file name. In many operating systems, this file name is a case sensitive. (i.e., UNIX).
- An example file name in UNIX is ‘`./bc180_file1`’.



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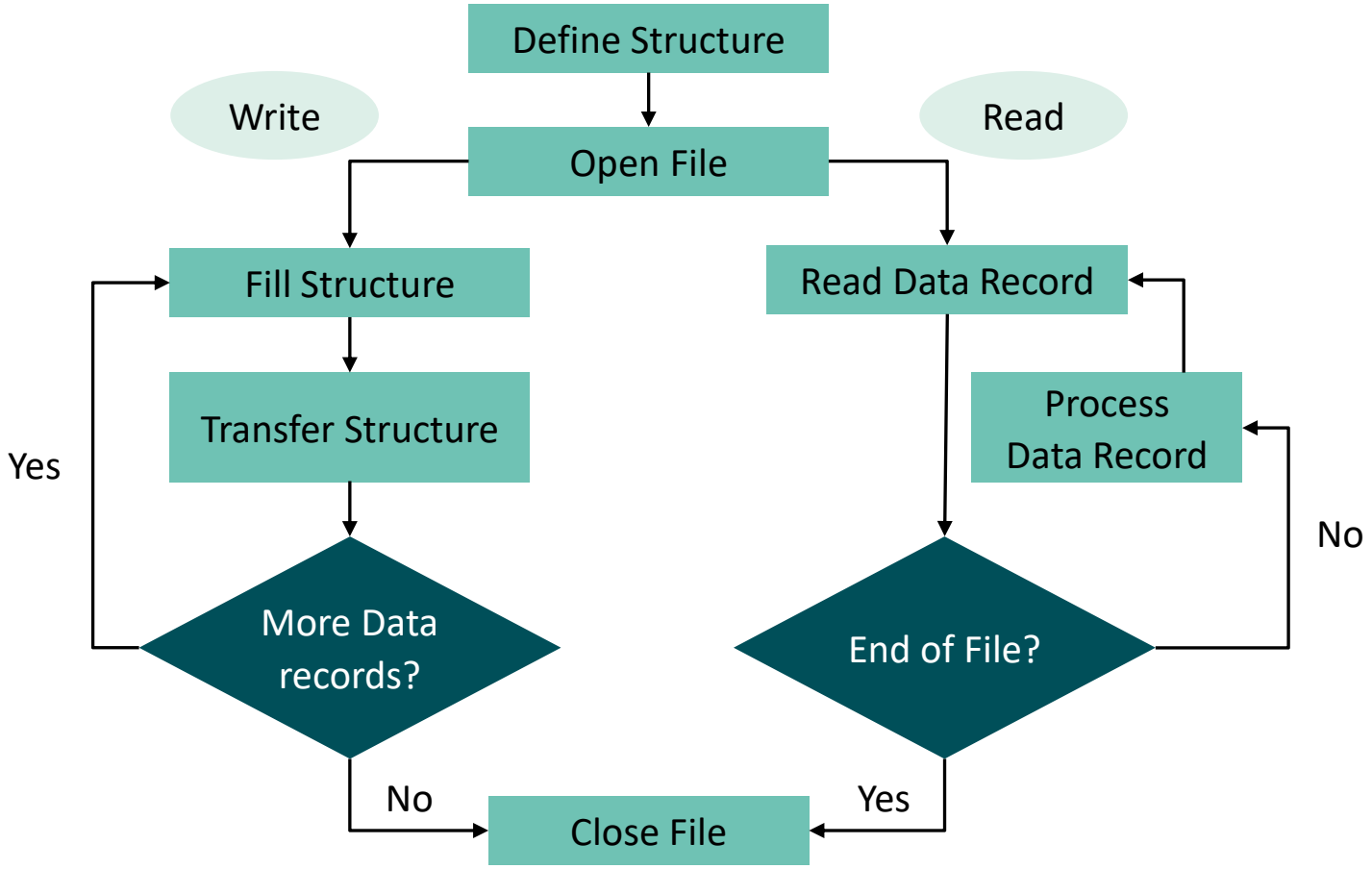
Overview Diagram—File Processing



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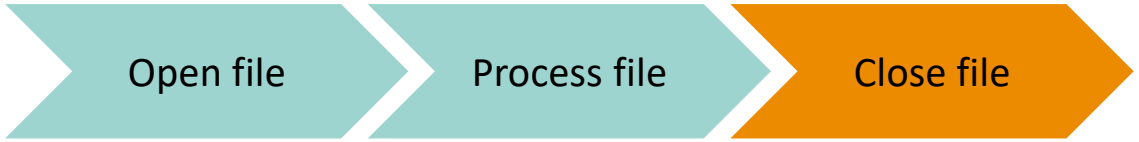
Close/Delete Files




```
CLOSE DATASET <filename>.
```

```
DELETE DATASET <filename>.
```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Maintain and Use Logical File Names: File—Transaction Code

Logical file name → **Physical file name**

In Customizing Assignment see transaction "file"

„Logicfile"
Logical file name

→ ***NT***

D:\usr\sap\trans\BC420_newfile_19990927.txt

→ ***UNIX***

/usr/sap/trans/BC420_newfile_19990927.txt

→ **...**

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Create Logical Files

Example for creating logical file is given below:

Log.file	LOGICFILE
Description	Sequential file with customers (external date)
Physical file	BC420_newfile_<DATE>.txt
Data format	ASC
Application area	PP
Logical path	tmp

Physical fire on operating system

The path can also be defined platform-independent beforehand

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File_Get_Name Function Mode

The file name format is dependent on the operating system. The function module File_GET_NAME is used to get platform-independent file names that can be used in application programs. This function module provides the physical file name associated with a specified logical file name.

You can find a description of the function module in the documentation for the function builder. (Library of function modules)

```
DATA: file like filename-fileintern.
…
CALL FUNCTION ‘FILE_GET_NAME;
     EXPORTING
         logical_filename            = ‘LOGICFILE’
     IMPORTING
         FILE_NAME                   = file
     EXCEPTIONS
         FILE_NOT_FOUND              = 1
         OTHERS                      = 2
IF sy-subrc <> 0.
  Write: / ‘file not found’. EXIT.
ELSE.
  Write:/ ‘physical file is’. file.
ENDIF.
…
```

The logical file is an import parameter of the function module

Depending on the operating system the physical file name is returned

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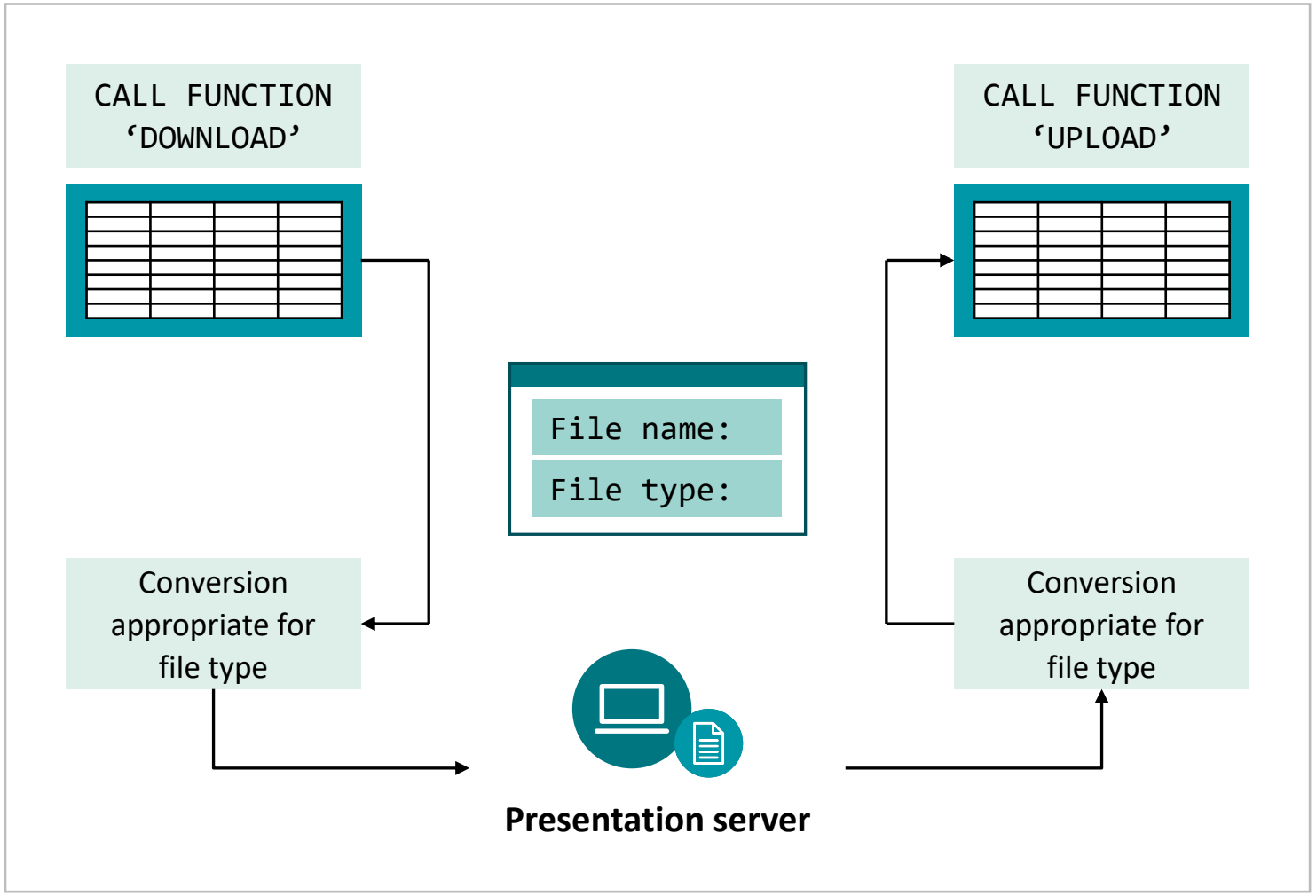
Local Sequential Files

To work with files on the presentation server, we use function modules `GUI_DOWNLOAD` and `GUI_UPLOAD`



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Keywords

Open Dataset	For opening files
Read Dataset	To read data from a file on the application server
Transfer	To write data to a file on the application server
Close Dataset	For closing files
Delete Dataset	For deleting files
Input	Reading the file data
Output	Writing to file
For Appending	To open a file, so that you can append data to the end of it
Gui_upload	Upload data from pc to the internal table
Gui_download	Download an internal table to the pc

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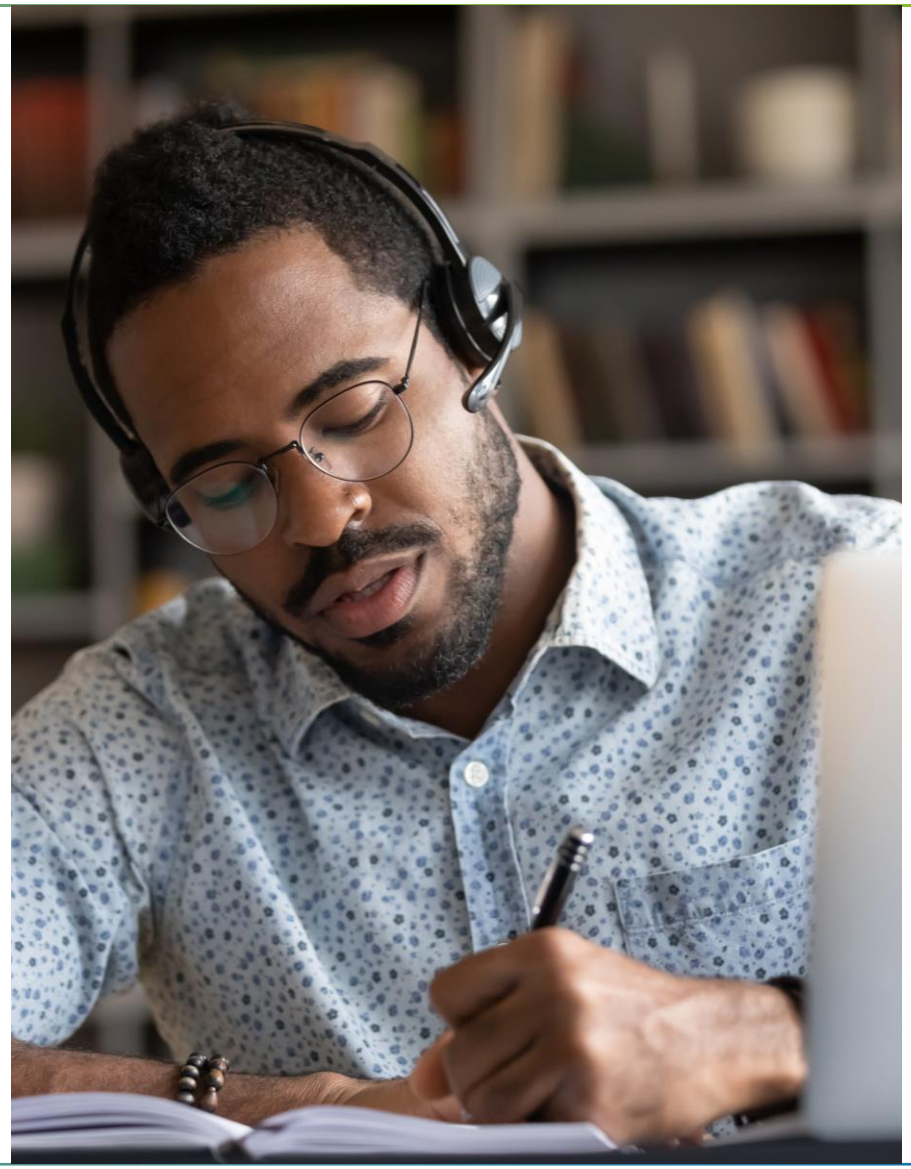
Hands-On Labs



Hands-On Activity 1

Activity Details	
Problem Statement	Perform the exercises in the ABAP Conversions: Exercises document

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Summary

Here are the key learning points of the module.

- Master data and transaction data are the two types of data. Conversion and interface are the types of data transfer.
- Data integrity must be guaranteed when transforming data from another SAP system or from an external system.
- Data analysis specifies how existing data should be mapped to SAP data structures.
- The SAP system comprises of three distinct layers—Database, application, and presentation.
- File processing refers to either creating a file or reading a file.
- The File_GET_NAME function module provides the physical file name associated with a specified logical file name.

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